

INTERNSHIP PROJECT ASSIGNMENTS

Data Science / Machine Learning — Week 1
Individual Dataset Assignments

Assignment Summary

Student	Dataset	Project Title
Mugesh M	product_sales_dataset.csv	Product Sales Analysis
Muthusankar V	StudentPerformanceFactors.csv	Student Performance Factor Analysis
Lakshana	student_placement_career_success_dataset_2026.csv	Student Placement & Career Success Analysis
Palani Bharathi M	loan_data_new.csv	Loan Default Risk Analysis
Ramkumar S	03_FIDE_Chess_Ratings.csv	FIDE Chess Ratings Analysis
Nithish T	global_poverty_economic_inequality.csv	Global Poverty & Economic Inequality Analysis
Udayanidhi S	Gold_vs_Economic_Factors_2015_2026.csv	Gold Price vs Economic Factors Analysis
Amanullah Fazil S	global_hantavirus_surveillance_dataset_2026.csv	Global Hantavirus Surveillance — Epidemiological Analysis
Niteshvarshan	bitcoin_dataset.csv	Bitcoin Price Time Series Analysis

Mugesh M

Project: Product Sales Analysis
Dataset: [product_sales_dataset.csv](#)

Objective

Analyze product sales data to uncover top-performing categories, cities, and time-based trends.

Tasks

- Load and clean data — check for nulls, fix date formats, remove duplicates
- Find top 5 best-selling products and categories by Total_Sales_USD
- City-wise sales breakdown — which city generates the most revenue
- Monthly sales trend using line chart (parse Order_Date properly)
- Price vs Quantity correlation using scatter plot
- Final insight: 3 business recommendations based on your findings

Required Charts

- Bar chart — Top 5 products by sales
- Pie chart — Category-wise revenue share
- Line chart — Monthly sales trend
- Scatter plot — Price vs Quantity Sold

Deliverable

Jupyter Notebook with complete EDA, 4 labeled charts, observations per chart, and a business recommendations section

Muthusankar V

Project: Student Performance Factor Analysis

Dataset: [StudentPerformanceFactors.csv](#)

Objective

Identify which factors such as sleep, attendance, tutoring, and family income most impact student exam scores.

Tasks

- Load and explore data — check all columns, data types, and null values
- Encode categorical columns (Low/Medium/High) using map or LabelEncoder
- Correlation heatmap — identify top 5 features most correlated with Exam_Score
- Compare exam scores across School_Type (Public vs Private)
- Analyze the effect of Parental_Involvement and Internet_Access on scores
- Bar chart — average exam scores grouped by Motivation_Level
- Conclusion: which 3 factors matter most and provide reasoning

Required Charts

- Heatmap — feature correlations with Exam Score

- Box plot — scores by School Type
- Bar chart — average score by Motivation Level
- Bar chart — Internet Access vs average score

Deliverable

Jupyter Notebook with full EDA, encoded data, 4 charts, and a written conclusion

Lakshana

Project: Student Placement & Career Success Analysis

Dataset: [student_placement_career_success_dataset_2026.csv](#)

Objective

Determine which skills and profile attributes lead to higher placement rates and better salary packages.

Tasks

- Load and clean data — handle nulls and understand all column definitions
- Placement rate comparison across College_Tier levels
- CGPA vs Salary_LPA scatter plot — evaluate the impact of academic performance
- Compare DSA_Problems_Solved and LeetCode_Rating between placed and unplaced students
- Identify which Specialization yields the highest average salary
- Internship count vs placement status — assess the value of practical experience
- Top 5 features most correlated with Salary_LPA

Required Charts

- Bar chart — placement percentage by College Tier
- Scatter — CGPA vs Salary
- Box plot — LeetCode Rating: placed vs not placed
- Bar chart — average salary by Specialization

Deliverable

Jupyter Notebook with EDA, 4 charts, and key findings on placement readiness factors

Palani Bharathi M

Project: Loan Default Risk Analysis

Dataset: [loan_data_new.csv](#)

Objective

Analyze which factors — income, credit score, loan amount — are most associated with loan default.

Tasks

- Load and clean data — handle nulls and correct column naming inconsistencies
- Default rate breakdown by Education level and Gender
- Income distribution comparison for defaulters vs non-defaulters using histogram
- Credit Score vs Loan Status — box plot comparison
- Loan Amount vs Loan Interest Rate scatter plot, colored by default status
- Correlation heatmap of all numerical features
- Conclusion: identify the top 3 risk indicators for loan default

Required Charts

- Histogram — income distribution: defaulters vs non-defaulters
- Box plot — Credit Score by Loan Status
- Scatter — Loan Amount vs Interest Rate (colored by status)
- Heatmap — numerical feature correlations

Deliverable

Jupyter Notebook with risk analysis, 4 charts, and a top-3 risk factor summary

Ramkumar S

Project: FIDE Chess Ratings Analysis

Dataset: [03_FIDE_Chess_Ratings.csv](#)

Objective

Explore the global distribution of chess ratings across federations, genders, and player titles.

Tasks

- Load and clean data — handle NA titles, check for nulls and data types
- Top 10 federations by average player rating
- Rating distribution by gender — histogram and box plot comparison
- Title distribution — count of players holding each title (GM, IM, FM, etc.)
- Birth year vs Rating scatter plot — analyze the relationship between age and rating
- Top 10 highest rated players displayed as a formatted table
- K-factor analysis — distribution of K values and their significance

Required Charts

- Bar chart — top 10 federations by average rating

- Box plot — rating distribution by gender
- Bar chart — player title distribution
- Scatter — birth year vs rating

Deliverable

Jupyter Notebook with full EDA, 4 charts, and a written summary on rating patterns

Nithish T

Project: Global Poverty & Economic Inequality Analysis

Dataset: [global_poverty_economic_inequality.csv](#)

Objective

Analyze global poverty trends, GINI inequality, and how GDP per capita and HDI relate to poverty rates across regions.

Tasks

- Load and clean all 25 columns — parse data types and handle missing values properly
- Top 10 countries by poverty rate vs top 10 by GDP per capita
- GINI coefficient distribution by region — identify the most unequal region
- HDI vs Poverty Rate scatter plot — examine if higher HDI consistently reduces poverty
- Correlation between literacy rate, life expectancy, and poverty rate
- Time series — average global poverty rate trend across available years
- Income group comparison — all key indicators side by side

Required Charts

- Bar chart — top 10 highest poverty rate countries
- Scatter — HDI vs Poverty Rate
- Box plot — GINI coefficient by region
- Line chart — global poverty trend over years

Deliverable

Jupyter Notebook with multi-dimensional analysis, 4 charts, and a global inequality summary

Udayanidhi S

Project: Gold Price vs Economic Factors Analysis

Dataset: [Gold_vs_Economic_Factors_2015_2026.csv](#)

Objective

Analyze how gold prices have been influenced by the USD index, crude oil prices, and inflation from 2015 to 2026.

Tasks

- Load data and parse the Date column properly as a datetime index
- Plot the full gold price trend from 2015 to 2026 as a line chart
- Correlation matrix — gold price vs USD index, crude oil, and inflation
- Year-wise average comparison of all four variables
- Identify periods of significant gold price spikes and overlay with inflation data
- Compute and plot 30-day rolling average of gold price using pandas rolling()
- Conclusion: identify the factor with the strongest influence on gold price

Required Charts

- Line chart — gold price trend 2015–2026
- Heatmap — correlation of all 4 economic variables
- Multi-line chart — all variables normalized to same scale
- Bar chart — year-wise average gold price

Deliverable

Jupyter Notebook with time-series analysis, rolling averages, 4 charts, and an economic interpretation

Amanullah Fazil S

Project: Global Hantavirus Surveillance — Epidemiological Analysis

Dataset: [global_hantavirus_surveillance_dataset_2026.csv](#)

Objective

Conduct a comprehensive epidemiological analysis of global hantavirus cases — covering transmission patterns, fatality factors, environmental correlations, and outbreak feature profiling.

Tasks

- Load and clean all 19 columns — parse dates, handle nulls, encode all categorical variables
- Case distribution by country and region — identify outbreak concentration zones
- Fatality rate by virus strain — determine which strain has the highest mortality
- Transmission type analysis — breakdown by type and exposure source
- Environmental correlation — analyze relationship between temperature, humidity, rodent presence index and fatality outcome
- Age group analysis — create age bins and identify the most affected demographic
- Recovery days comparison — hospitalized vs non-hospitalized patients
- Feature importance chart — identify top predictors of fatality using correlation or a basic model

- Time series — monthly case count trend derived from report_date
- Final epidemiology report section: 5 key findings with public health recommendations

Required Charts

- Bar chart — case count by country
- Heatmap — environmental factors vs fatality
- Box plot — recovery days by hospitalization status
- Line chart — monthly case trend
- Bar chart — fatality rate by virus strain

Deliverable

Complete Jupyter Notebook with 5+ charts, encoded features, correlation analysis, feature importance visualization, and a written Epidemiology Report section

Niteshvarshan

Project: Bitcoin Price Time Series Analysis

Dataset: [bitcoin_dataset.csv](#)

Objective

Analyze Bitcoin historical OHLCV data to identify price trends, volatility patterns, volume behavior, and moving averages.

Tasks

- Load and parse the Date column as a datetime index
- Plot the full Bitcoin closing price history as a line chart
- Calculate and overlay 7-day, 30-day, and 90-day rolling averages on the same chart
- Compute daily return: $(\text{Close} - \text{Open}) / \text{Open} * 100$ — add as a new column
- Volatility analysis — identify the top 10 most volatile trading days
- Volume vs price movement — examine whether high volume correlates with large price changes
- Year-wise average closing price — identify years with the highest growth
- Identify and annotate bull and bear market periods from the dataset

Required Charts

- Line chart — full price history with 7/30/90-day rolling averages
- Bar chart — year-wise average closing price
- Histogram — daily return distribution
- Scatter — trading volume vs daily price change

Deliverable

Jupyter Notebook with time-series analysis, rolling averages, volatility study, 4 charts, and a market cycle summary

General Submission Guidelines

- Deadline: End of Week 1 — Friday
- Submit Jupyter Notebook (.ipynb) via Google Classroom or the shared link provided
- File naming format: YourName_ProjectName_Week1 (e.g., Mugesh_ProductSales_Week1)
- All charts must include proper titles, axis labels, and a consistent color theme
- Write at least one to two lines of observation below each chart in a markdown cell
- The final cell must be a markdown summary containing key findings from the analysis

Work independently. Approach your mentor only after making two genuine attempts to resolve the issue on your own.